

CS 340 README Template

About the Project/Project Title

This project is a web-based dashboard created for Grazioso Salvare using Python, MongoDB, and the Dash framework. The dashboard connects to a MongoDB database containing Austin Animal Center for animal outcome records and displays the data in an interactive table. Users can sort, filter, and browse animal records while viewing a geolocation map and data visualization chart.

Motivation

The purpose of this project is to help Grazioso Salvare identify dogs that may be suitable for rescue and training programs. The dashboard allows users to quickly access animal data, visualize information, and view animal locations on a map. The project demonstrates how a MongoDB database can be integrated with a Python web application to create an interactive data analysis tool.

Getting Started

1. Start the MongoDB service.
2. Ensure the AAC database and animals collection are loaded with the Austin Animal Center Outcomes dataset.
3. Place the following files in the same project folder:
4. ProjectTwoDashboard.ipynb
5. CRUD_Python_Module.py
6. Grazioso Salvare Logo.png
7. Open Jupyter Lab.
8. Run all cells in the notebook.
9. Open the dashboard link generated by Dash.

Installation

- Python 3
- MongoDB
- Jupyter Lab
- Dash
- Dash Leaflet
- Pandas
- NumPy
- Matplotlib
- Plotly
- PyMongo

Usage

Note: This template has been adapted from the following sample templates: [Make a README](#), [Best README Template](#), and [A Beginners Guide to Writing a Kickass README](#).

Use this space to show useful examples of how your project works and how it can be used. Be sure to include examples of your code, tests, and screenshots.

Code Example

Example of retrieving all records from MongoDB:

```
from CRUD_Python_Module import AnimalShelter
```

```
username = "aacuser"
```

```
password = "LeBron"
```

```
shelter = AnimalShelter(username, password)
```

```
df = pd.DataFrame.from_records(shelter.read({}))
```

Example of generating the breed chart:

```
breed_counts = dff['breed'].value_counts().head(10)
```

```
fig = px.bar(  
    x=breed_counts.index,  
    y=breed_counts.values,  
    title="Top 10 Animal Breeds"  
)
```

Example of displaying the selected animal on the map:

```
dl.Marker(  
    position=[dff.iloc[row,13], dff.iloc[row,14]]  
)
```

Tests

Database connection test:

Verified that the dashboard successfully connects to MongoDB using the AnimalShelter CRUD module.

Screenshots

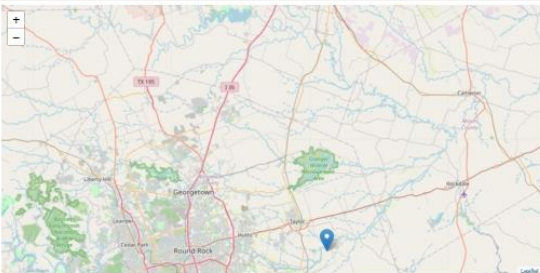
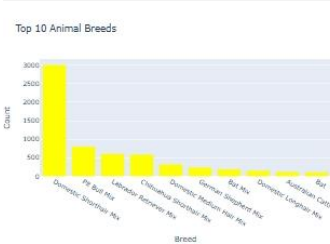
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SNHU CS-340 Dashboard

Created by Jose Alonso



	rec_num	age Upon Outcome	animal_id	animal_type	breed	color	date of birth	datetime	monthyear	name	outcome_subtype	outcome_type	sex Upon Outcome	location_lat	location_lng
Filter data...															
1	1	3 years	A746874	Cat	Domestic Shorthair Mix	Black/White	2014-04-10	2017-04-11 09:00:00	2017-04-11/09:00:00		SNP	Transfer	Neutered Male	30.5866578739495	-97.348878872258
2	2	1 year	A725717	Cat	Domestic Shorthair Mix	Silver Tabby	2015-05-02	2016-05-06 10:49:00	2016-05-06/10:49:00		SNP	Transfer	Spayed Female	30.6121084560228	-97.742593674644
3	3	2 years	A716338	Dog	Chihuahua Shorthair Mix	Brown/White	2013-11-18	2015-12-28 18:43:00	2015-12-28/18:43:00	Fram		Adoption	Neutered Male	30.7091748121648	-97.552173388751
4	4	7 months	A713653	Cat	Siamese Mix	Seal Point	2016-01-25	2016-08-27 18:11:00	2016-08-27/18:11:00	Kitty		Adoption	Intact Female	30.1188863374257	-97.7248176793891
5	5	2 years	A695584	Dog	Labrador Retriever Mix	Tan/White	2012-11-06	2015-05-30 13:48:00	2015-05-30/13:48:00	Luke		Return to Owner	Neutered Male	30.7184815618433	-97.54220743528
6	6	5 years	A698064	Dog	Cardigan Welsh Corgi Mix	White/White	2010-01-27	2015-01-28 10:39:00	2015-01-28/10:39:00	Lacy	Rabies Risk	Euthanasia	Spayed Female	30.4273165854231	-97.78797152966
7	7	2 years	A673838	Dog	Pit Bull Mix	Black/White	2012-03-03	2014-03-19 15:15:00	2014-03-19/15:15:00	*Seth	Aggressive	Euthanasia	Neutered Male	30.29542516583441	-97.3136542118438
8	8	1 year	A735551	Dog	Labrador Retriever/Australian Cattle Dog	Black	2015-10-12	2016-11-27 18:08:00	2016-11-27/18:08:00	*Mia		Adoption	Spayed Female	30.4441212820362	-97.732058833879
9	9	3 years	A726214	Dog	Labrador Retriever Mix	Red/White	2013-02-04	2016-02-11 12:41:00	2016-02-11/12:41:00	Blissling		Adoption	Spayed Female	30.3878848109411	-97.368433077137
10	10	3 months	A684208	Cat	Domestic Shorthair Mix	Tortie	2013-09-01	2013-12-08 14:58:00	2013-12-08/14:58:00	*Taylor		Adoption	Spayed Female	30.7581185481848	-97.51820210884



Contact
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